A Blockchain-Based Network Security Mechanism for Voting Systems

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***Abstract*—As technology advances, many countries have now opted for electronic voting systems. Any voting system must follow principles of transparency and impartiality in order to achieve fairness; the electronic voting process must also be protected against cyberattacks or denial-of-service attacks (DDOS) because such attacks may affect the processing time in voting procedures and even hinder the fairness in voting. This study establishes a network security mechanism for voting systems based on blockchain technology. The blockchain mechanism employs a distributed architecture that can prevent system shutdown resulting from malicious cyberattacks; additionally, any user in the blockchain can authenticate data integrity, which satisfies requirements of transparency and impartiality in voting systems. This study utilizes bilinear pairing to establish network security in voting systems, which call for anonymity, authenticity, integrity, and non-repudiation. When authenticating voting integrity, the user's anonymity must be ensured to prevent identity revelation, and the data must be protected against malicious tampering such security measures also fulfill blockchain requirements. The proposed voting system relies on the basis of blockchains to create a trustworthy voting system. In current blockchain technology, smart contracts allow the establishment of voter-related regulations to prevent controversies during voting processes. Additionally, in order to establish both a secret ballot and an open ballot system, the study also implements a bilinear pairing security mechanism to ensure the overall security of a voting procedure.**

***Index Terms*—BlockChain, Voting Mechanism, Network Security, Bilinear Pairing**

I. INTRODUCTION

Many countries nowadays practice voting methods. However, traditional voting mechanisms are limited by factors such as distance of polling places, voter registration area, or whether the voter presents adequate identification, all of which might affect a voter's voting rights. If the voting method requires manual operation, controversies or abuse might ensue. Hence, many countries or states have developed electronic voting systems [1], such as California and Australia, and have added cryptography technology to their electronic voting to enhance the voting system's security and credibility. Aside from minimizing human error, electronic voting allows for quick data analysis. Currently, many electronic voting systems employ centralized servers for data processing; nevertheless, centralized servers are susceptible to malicious attacks and DDOS attacks, so distributed architecture should be used instead in order to reduce the risk of post-attack shutdown.

Blockchain-related researches emerged mainly after Bitcoin was developed. Blockchains mainly employ a distributed storage mechanism; the decentralized system allows users to trade freely while verifying data accuracy. Blockchain 1.0 focused on digital currency; Blockchain 2.0 emphasized smart assets development; Blockchain 3.0 mostly concerned smart contracts. Smart contracts drastically reduces the cost of contract negotiation , contract execution, and payment issues. Autonomous agents eliminate the cost of agencies and negotiations and might even lead to a highly distributed enterprise that requires none to only the bare minimum management. Smart contracts speed up contract negotiation processes and even transaction speed in a sharing economy.

This study has developed a voting system with network security mechanism using blockchain technology. The blockchains adopt a distributed ledgers model that allow for any form of transaction via P2P architecture; every transaction's details are stored in each node so the blockchains do not require a centralized database for transaction data storage, and the blockchains only allow addition of information and not modification of original data. The blockchains enjoy the following features: (1) they adopt a distributed architecture that store data in different nodes; (2) all the nodes jointly maintain the public ledger so that if any node's ledger is ever lost, other nodes can still replicate and use it; and (3) the blockchains employ cryptography techniques including hash technology and elliptic curve cryptography to ensure network security. Our proposed blockchain-based voting system not only prevents system shutdown, but also enables any user to conduct voting authentication.

The study utilizes bilinear pairing to establish a voting system's network security. Given that when it comes to network security, blockchains lack anonymity while many voting systems demand it and also require authentication of voter identity, we have enabled all network users to engage in voting authentication. All voting results are distributed in each nod while each node can record ledgers of each election so that when related departments wish to acquire election results, they can retrieve information from ledgers. This study

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features bilinear pairing for network security mechanism. The anonymous voting mechanism ensures the identity and privacy of voters while the identity authentication mechanism not only protects voter identity but also prevents cases of repeated voting. The bilinear pairing technology is based on elliptic curve cryptography, which ensures network security. This study also conducted feasibility analysis and performance analysis based on the experiment results, which prove our scheme's feasibility.

II. RELATED WORKS

The social networking voting mentioned in this study [2] utilizes matrix factorization and nearest-neighbor techniques to establish a voting system with enhanced voting accuracy. The system applies matrix factorization for its online electronic voting and integrates a recommender system to explore marketing research such as commodities on social networking sites. This study's [3] applies voting systems to message authenticity in Internet of Vehicles: messages passing on the network are calculated using voting between vehicles to determine message authenticity. This study's [4] proposes a distributed multi-choice voting/ranking algorithm to calculate voting result ranking and relevant data. Our proposed scheme can effectively reduce time complexity in counting ballots. This study's [5] mentions the cyberattack incident revealed in the 2016 United States presidential election; meanwhile, the traditional voting mechanism also showed troubles of human error instances, and all these affect an election's fairness. Hence, it is imperative to replace traditional voting with electronic voting and enhance network security in electronic voting so as to ensure election fairness. This study proposes a voting system network security mechanism based on blockchain technology that improves transparency and impartiality in electronic voting.

This study [6] proposes a weighted voting system. It utilizes several states, each one weighted, so that when the weighted value exceeds the default value, the system deems the voting a failure. This study [6] focuses on establishing a dependable voting system. This study [7] suggests that voting advice applications are mainly applied in social voting advice application. This study [7] helps gather information on each voting district electorate's opinions and political outlooks. This study [8] proves that E2E voting systems can indeed be subject to fairness and security verification; hence, the voting systems can be transferred to anonymous voting systems in class.

III. THE PROPOSED SCHEME

A. System Model

Figure 1 illustrates our proposed system model. As shown in Figure 1, many identity authentication units focus on authenticating the voter's identity. When a voter is successfully authenticated and completes their voting, the system places the voting result on the node so that the node may authenticate the effectiveness of the voting outcome while other users can authenticate the effectiveness and integrity of every piece of data on the node, thus achieving election transparency and fairness.

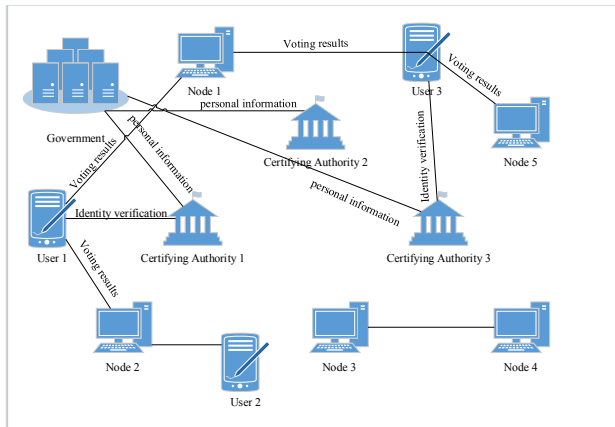

Fig. 1. System Illustration.

B. Voter Authentication

The notation used in this study are shown in Table I. Suppose User1 ($\mathbb{ID}_{t,u_1}$) wishes to conduct voting. First, they must undergo identity authentication with Certifying Authority 1 ($\mathbb{ID}_{t,CA_1}$) , which has already obtained the user's identity information from the government. When $\mathbb{ID}_{t,u_1}$ uses $\mathbb{ID}_{t,CA_1}$ public key to encrypt personal information, and then transmit it to $\mathbb{ID}_{t,CA_1}$ for authentication. $\mathbb{ID}_{t,CA_1}$ creates $\mathbb{ID}_{t,u_1}$'s ballot information (M), anonymous ID ($\mathbb{ID}_{p,u_1}$), public key ($\mathcal{PK}_{\mathbb{ID}_{p,u_1}}$), and private key ($\mathcal{PR}_{\mathbb{ID}_{p,u_1}}$). Meanwhile, $\mathbb{ID}_{p,u_1}$=($\mathbb{ID}_{p,u_1}||r$), $\mathcal{PK}_{\mathbb{ID}_{p,u_1}}$=$(\mathbb{ID}_{p,u_1} \cdot T \cdot \mathbb{ID}_{t,CA_1} \cdot P)^r$, and $\mathcal{PR}_{\mathbb{ID}_{p,u_1}}$=$(\mathbb{ID}_{p,u_1} \cdot T \cdot \mathbb{ID}_{t,CA_1} \cdot P)^r \cdot s$. $\mathbb{ID}_{t,CA_1}$ uses $\mathbb{ID}_{t,u_1}$'s private key to encrypt the information, and then transmit it to $\mathbb{ID}_{t,u_1}$; $\mathbb{ID}_{t,u_1}$ in turn uses the private key for decryption. The process is as follows:

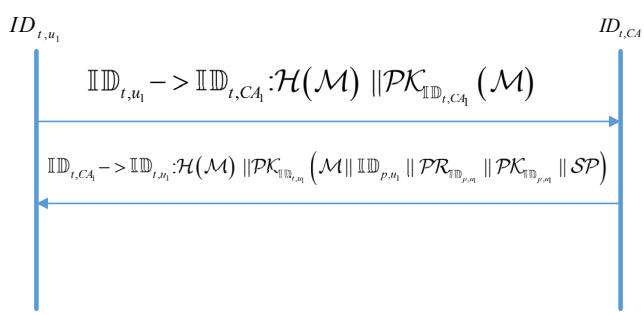

Fig. 2. Illustration of Voter Authentication.

C. Data Transmission and Authentication

When $\mathbb{ID}_{p,u_1}$ successfully completes identity authentication, $\mathbb{ID}_{p,u_1}$ uses the public key to encrypt information such as voting result and transmit it to the node. The node then authenticates whether $\mathcal{PK}_{\mathbb{ID}_{p,u_1}}$ was indeed sent by $\mathbb{ID}_{t,CA_1}$; if

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TABLE I
NOTATION TABLE

notation	content
$\mathbb{ID}_{t,U}$	The real ID of the user U.
$\mathbb{ID}_{p,U}$	The anonymous ID of the user U.
G_1	The additive group.
P	The generator of G_1.
H	The hash function.
M	Message
T	Timestamp
$\mathcal{PK}_U$	The public key of user U.
$\mathcal{PR}_U$	The private key of user U.
$\mathcal{SP}$	The public value of user U.

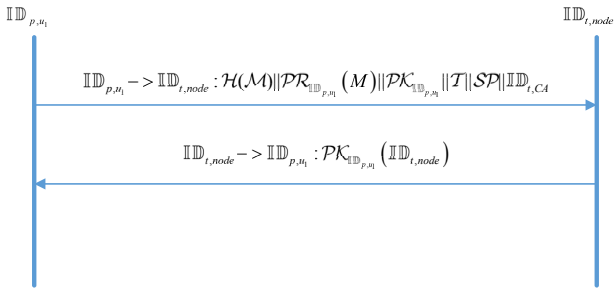

Fig. 3. Illustration of Voter Authentication.

yes, then the deciphered text shall reveal $\mathbb{ID}_{p,u_1}$ as a legitimate user. The process is as follows:

When another user wishes to authenticate $\mathbb{ID}_{p,u_1}$'s voting results, they must first authenticate whether the public key was indeed issued by $\mathbb{ID}_{t,CA_1}$ by $e(\mathcal{PK}_{\mathbb{ID}_{p,u_1}}, \mathcal{SP})$=$e((\mathbb{ID}_{p,u_1} \cdot T \cdot \mathbb{ID}_{t,CA_1} \cdot P)^r, \frac{1}{r} \cdot P)$=$e(\mathbb{ID}_{p,u_1} \cdot T \cdot \mathbb{ID}_{t,CA_1} \cdot P)$, in which $\mathbb{ID}_{p,u_1}$, $\mathbb{ID}_{t,CA_1}$, and T are public information. If the computed values are identical, this signifies that $\mathbb{ID}_{p,u_1}$ is a legitimate user, and other users may trust $\mathbb{ID}_{p,u_1}$'s voting result.

IV. PERFORMANCE ANALYSIS

A. Security Analysis

Our proposed network security measure focuses on analysis of anonymity, authenticity, integrity, and non-repudiation. The cryptography techniques used in this study apply bilinear pairing [9]. The following are the network security analysis:

1) Anonymity: under our proposed scheme, prior to voting, each user must undergo identity authentication. $\mathbb{ID}_{t,CA_1}$ assigns the user an anonymous ID so that other users cannot deduce their true ID from their anonymous ID.
2) Authenticity: the public key is computed using the user's ID so as to authenticate the user's true identity.
3) Integrity: hash technology is employed for authenticating message integrity to prevent the messages from being maliciously manipulated.
4) Non-repudiation: all parties must use public/private key for encryption and decryption, and since the public/private key is obtained using a party's ID, the origin of message is ensured.

B. Performance Analysis

This study utilizes the cryptography technique performance analysis from References [10], [11], [12] to further investigate the method proposed in this study, shown in Table II, while the method's performance analysis is shown in Table III. Table III demonstrates that our proposed method bears low computational burden, provides anonymity, and can be implemented in voting systems to allow other users for authentication.

TABLE II
EXECUTION TIME IN MILLISECONDS.

Notation	Description	Execution time (ms)
TP	Pairing operation	$\approx$ 4.5
TM	Point Multiplication	$\approx$ 0.6
TE	Field Exponentiation	$\approx$ 0.45
H	HMAC	0.002
ASE	Asymmetric Encryption	0.19
ASD	Asymmetric Decryption	4.65

TABLE III
EFFECTIVENESS ANALYSIS.

Property	Spending time	Quantity of data
Identity Authentication Encryption	$H + ASE$ =0.192(ms)	320 byte
Identity Authentication Decryption	$H + ASD$ =4.652(ms)	320 byte
Massage Communication Encryption	$H + ASE$ =0.192(ms)	960 byte
Massage Communication Decryption	$H + ASD$ =4.652(ms)	960 byte
Massage Authentication	$n * (H + ASD)$	n*960 byte

V. CONCLUSION

This study has successfully developed a blockchain-based network security mechanism for voting systems. Given that blockchains feature a distributed architecture and only allow addition but not modification of data, this effectively allows any user to conduct data authentication. We have also employed bilinear pairing in establishing the network security mechanism; bilinear pairing has the advantage of low encryption/decryption data volume, which reduces volume of stored data in nodes. Additionally, we have achieved the goal of anonymity and provided an anonymous voting system. In the future, we expect to add mechanisms such as ballot counting and ranking to our scheme and improve the speed of vote counting and complement electronic voting systems. In the future, this study's proposed voting scheme can be tested through simulation for feasibility; the study also plans to establish a comprehensive voting system and work on improving the speed of blockchain authentication in order to expedite functions such as vote counting and ranking.

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